

ARE SMART HOMES MORE EFFICIENT?

A residential automated demand response program works on reducing California's peak time loads.

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For nearly 25 years, utilities have been trying out various programs designed to get customers to reduce their loads during times when power is scarce. These demand response (DR) programs, which typically use either direct messaging or pricing signals to alert customers to power scarcities, can bring substantial benefits. Utilities can “purchase” load from their customers for lower rates than they would pay to provide it, lowering the utility system’s peak demand and helping to reduce peak wholesale power market prices. In addition, customers can be asked to reduce load during nonpeak periods to help maintain grid reliability, defer or eliminate generation capacity expansion, or defer or eliminate transmission/distribution capacity expansion. Now, new smart meters and gateway systems can make such demand response in homes and small businesses convenient, effective, and economical.

The California Energy Commission (CEC) hired Rocky Mountain Institute (RMI) to assess the impact of one such program—the Automated Demand Response System (ADRS) pilot program—on the load of residential customers. In particular, CEC asked RMI to estimate the average ADRS residential customer’s load response and to evaluate whether the level of load impacts of ADRS residential participants has increased, decreased, or stayed the same over two summers in response to price signals, after controlling for weather and other independent factors. The pilot



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California single-family homes can benefit from the convenience and efficiency of demand response programs, such as the ADRS pilot program.

program was conducted jointly by three utilities: Pacific Gas & Electric (PG&E), Southern California Edison (SCE), and San Diego Gas & Electric (SDG&E).

ADRS in California

The ADRS pilot participants were first recruited in 2004 from owner-occupied, single-family homes located in California’s Statewide Pricing Pilot climate zone 3 (hot inland, according to the ADRS pilot program’s climate zoning conventions), in zip codes served by television cable providers who agreed to cooperate in the pilot program. Customer recruitment was concentrated in certain zip codes: PG&E homes were in Woodland and Stockton; SCE homes were in Valencia, Santa Clarita, Saugus, and Los Angeles; and SDG&E homes were in San Diego. ADRS homes were recruited at random, although all homes had central air conditioning.

A total of 175 homes were initially recruited into the ADRS pilot program in 2004. Of these, 75 were from PG&E, 76 were from SCE, and 24 were from SDG&E. However, by July 2005, the start of the second year of the pilot, 44 participants had opted out of the program for various reasons, and only 131 homes remained. A set of 154 control homes was also identified from which the ADRS participant reductions were measured. Of the homes that completed the program, 35 were from PG&E, 58 from SCE, and 25 from SDG&E.

ADRS Technology

All ADRS customers were put on a dynamic pricing tariff, a time-dependent electric rate schedule called CPP-F. The homes were equipped with GoodWatts, an always-on, two-way communicating, advanced home climate control system with Web-based programming of user preferences for control of home ap-

pliances. The GoodWatts system is an Invensys Climate Controls product. Via the Internet, homeowners with GoodWatts can set climate control and pool or spa pump run-time preferences and view these settings at any time both locally and remotely. Participants can also view whole-house or end-use-specific demand in real time and display trends in historical consumption. The energy management technology includes the following components:

- wireless RF communications network connecting all system components;
- two-way communicating whole-house meter capable of recording consumption data in 15-minute intervals;
- wireless Internet gateway and cable modem;
- programmable smart thermostats;
- load control and monitoring (LCM) device to manage selected loads (e.g., pool pump); and
- Web-enabled user interface and data management software.

GoodWatts lets users constantly view the current electricity price online or via the thermostat. It also allows users to program desired thermostat and pool/spa responses to changes in electricity price. For ADRS homes with pools and spas, supplemental LCMs were installed to garner additional demand reduction during utility-triggered curtailment events.

Control homes were on standard tiered rates and were not equipped with GoodWatts. However, these homes were matched as ADRS participants because they were also owner-occupied, single-family homes from the Statewide Pricing Pilot climate zone 3 with central A/C. All homes were classified into one of two consumption strata based on summer average daily energy consumption (ADU). Homes were classified as high consumption if their summer ADU was 24 kWh or greater and as low consumption if it was less.

Analyzing the Data

The three utilities provided 15-minute interval load data for ADRS and control homes for June 1 through September 30, 2004 and 2005. Hourly temperature data were collected for the same periods, by zip code, based on Invensys's weather subscription service. For each utility, average kW load for each interval was calculated by consumption stratum, for event and nonevent days. The CPP-F dynamic pricing tariff is a time-of-use (ToU) tariff that includes a critical peak pricing (CPP) element. Prices were higher between 2 pm and 7 pm (peak period) relative to other hours, which are charged a base rate. Very high, superpeak prices were imposed during the peak period on up to 15 days of the summer when electricity supply is anticipated to fall short of meeting demand. These superpeak days are also known as event days. The values were then used to construct daily average load curves for ADRS and control homes.

ADRS load savings, compared to the control group, were calculated for each 15-minute period by subtracting the adjusted average ADRS load from the corresponding average control home load, for each 15-minute data interval. ADRS load reductions relative to the control group were calculated for event and nonevent days, by utility and by consumption stratum. A statewide average was calculated from the weighted average of the utility results, indicating that 90% of the time, loads will be within limits.

Load Impact Results

Overall load reduction on event days was smaller in 2005 than it was in 2004. This can be attributed mostly to lower control home loads in 2005, rather than to reduced ADRS performance. Average superpeak period control home consumption in 2005 decreased by 8% compared to 2004, in spite of the fact that 2005 was hotter on average during the six summer months of the study. The lower average control home load in 2005 on event days is counterintuitive, and we cannot explain this finding with available data. Household level

investigation into control home consumption revealed a significant number of outliers, where control homes exhibited almost no energy usage throughout the entire day. These high-consumption control homes were removed from the sample for the 2005 analysis. The number of control homes removed did not reduce the statistical significance of 2005 results.

High-consumption ADRS loads, on the other hand, increased by 7% during superpeak periods on average in 2005, as was to be expected during months with more cooling degree-days. Note that the percent increase in ADRS superpeak period load is calculated from a lower overall peak period consumption than the overall peak period consumption.

Nonevent day performance of high-consumption ADRS homes exhibits similar patterns. For high-consumption ADRS customers, peak period load reduction was reduced by 0.86 kW or 32% relative to control homes in 2004, compared to 0.73 kW or 27% reduction in 2005, statewide. Unlike smaller peak period load reduction on event days, however, this smaller peak period load reduction on nonevent days is attributable mostly to higher average summer season temperatures in 2005. Both high-consumption ADRS and control customers had higher peak period demand in 2005. However, control load increased by 4% during the peak period in 2005, while ADRS load increased by 12% during that same period. Note also that ADRS loads dropped further at 2 pm in 2005 than it did in 2004, but recovered more quickly throughout the rest of the peak period. This finding is consistent with the observation that hotter summer weather causes the indoor temperatures to rise to the on-peak thermostat setpoint faster. This results in modestly diminished ADRS savings statewide.

Results also differed by utility. Load reduction of high-consumption ADRS homes within each utility service territory was substantial during both years, although performance was slightly better in 2004 than in 2005 for PG&E and SCE. SDG&E's very modest results can be attributed to the unusual behavior of the control group, as mentioned

above. SCE high-consumption ADRS customers achieved on average about 2 kW reductions on event days across a range of temperatures. PG&E and SDG&E high-consumption ADRS customers achieved substantial, but lower, reductions—close to 1 kW on event days on average. Load reduction impact of ADRS homes in SDG&E territory should be interpreted with caution, however, due to the small sample size. In SDG&E territory, just seven high-consumption homes participated in the program in 2004, and only six participated in 2005.

We cannot compare results across utilities, since each utility ran the ADRS pilot within its service territories independently of the other two. For this reason, there are numerous controllable and uncontrollable factors that would need to be held constant in order for a valid utility comparison to take place. However, one cannot help but wonder why SCE pilot performance was consistently stronger than pilot performance in PG&E and SDG&E territories. One factor that appears to provide a strong link to better performance in SCE territory is that ADRS homes were recruited from residential developments with homes that tended, on average, to be larger than ADRS homes in PG&E and SDG&E territory.

About 40% of SCE customers owned homes with floor areas larger than 2,000 ft² compared to about 30% and 20% for customers in PG&E and SDG&E service territories, respectively. Furthermore, the majority of ADRS participants (59%) in SCE territory had household incomes greater than \$100,000 per year. These homes also tended to have larger A/C units—on average 4 tons cooling capacity per unit—and were more likely to have additional controllable loads, such as swimming pools.

During 2005, the program recorded higher temperatures than it did during 2004, on both event and nonevent days. On both event and nonevent weekdays statewide, temperatures were nearly 8°F

warmer in 2005 during both July and August. September temperatures were 3°F warmer on event days. Average statewide nonevent day temperatures in September were essentially the same in 2004 and 2005.

Utility-specific temperatures exhibited similar patterns with the exception of PG&E, where average event day temperatures were higher in September 2004 than in September 2005. Both PG&E and SCE experienced average temperatures in the range of 89°F–98°F on event days and 80°F–95°F on non-

Both years show the same downward trend in load reduction over the duration of the superpeak period. In 2005, ADRS load reductions in PG&E territory fell more substantially between the first and last hours of the superpeak period than they did in 2004 (see Figure 1). For SCE service territory, however, high-consumption ADRS load reductions in both years were relatively constant during the first three hours of the superpeak period; they declined only during hours 4 and 5. Hourly percent reductions for ADRS homes in SDG&E territory declined steadily from hour 1 to hour 5, much as they did with PG&E. Relative performance between 2005 and 2004 was quite similar for the first four hours, with differences of only 5% on average.

Event day averages show that, due to load reductions during the superpeak period, ADRS high-consumption homes consumed less energy than control homes during the superpeak period in both 2004 and 2005 (see Figure 2). The shifting of load away from the superpeak period in ADRS homes is apparent in the relatively higher ADRS consumption in recovery and off-peak periods in 2005 and 2004. During the recovery period on event days (from 7 pm to 9 pm, the two hours immediately following the superpeak

period), ADRS customer consumption rebounded to exceed control consumption as ADRS thermostats were reset to off-peak period setpoints. ADRS homes also consumed more than control homes in the off-peak periods on event days in 2005. Off-peak period consumption on event days was the same in 2004, statewide, for ADRS homes and control homes. Nonevent day consumption patterns in 2005 and 2004 resemble the consumption patterns for event days. Differences between ADRS and control customers were more modest in 2004 statewide for recovery and off-peak periods. The 2005 nonevent day peak period reductions represent load shifting to off-peak periods rather than overall

Average Percent Reduction in Superpeak Consumption, PG&E High-Consumption Homes on Event Days

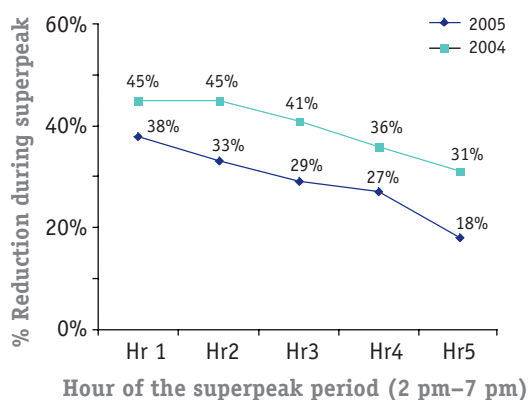


Figure 1. In 2005, ADRS load reductions in PG&E territory fell more substantially between the first and last hours of the superpeak period than they did in 2004.

event days. Temperatures in SDG&E territory, on the other hand, were on average 10°F cooler—between 76°F and 88°F on both event and nonevent days.

Percentage load reductions for high-consumption ADRS customers compared to control customers during each hour of the superpeak period between 2004 and 2005 were consistently less in 2005 than in 2004. As noted, this was mostly due to a lower relative 2005 control load during the peak period, resulting in lower savings in 2005. Nevertheless, superpeak performance in high consumption ADRS homes was reliable over the two consecutive years.

reduction in energy consumption over the whole day. On the other hand, both load shifting and energy conservation were present during 2004 nonevent weekdays.

Comparing daily energy consumption patterns between years by utility, high-consumption ADRS customers in PG&E and SCE service territory shifted load more aggressively from superpeak and peak periods to off-peak periods in 2005 than they did in 2004, with subsequent reductions in net energy conservation. High-consumption ADRS homes in SDG&E service territory, on the other hand, appeared to have used technology to reduce overall energy consumption as opposed to merely shifting load on both event and nonevent days, for both summer 2005 and summer 2004. It may be that in SDG&E, where average temperatures were typically 10°F cooler than the statewide average, customers were better able to respond to peak pricing signals by reducing energy consumption overall. In PG&E and SCE service territory, where temperatures tended to be higher than the statewide average, high-consumption ADRS customers resorted to shifting load in order to save money using automated technology.

In addition to load reduction from central air, the ADRS program also curtailed pool pump load for homes with swimming pools. In 2005, a total of 33 homes enrolled in the program had swimming pools. A typical pool pump operates continuously for four to eight hours each day. Owners typically schedule operation during daylight hours when the chlorine cycle is most efficient. High-consumption ADRS customers with swimming pools consistently scheduled pool pump operation outside of the hours between 2 pm and 7 pm, to reduce superpeak and peak period consumption every day. This rescheduling of pool pump operation contributed 32% of total superpeak

reduction for the average home with a pool. Since approximately one out of every three ADRS participants owns a pool, this load reduction contributed about 10% of total superpeak period reduction on event days. On nonevent days, rescheduling pool pump operation contributed over 50% of total peak period reduction for the average home with a pool. This load reduction contributed about 27% of total peak period reduction on nonevent days

target customers who are most likely to reduce substantial load. Programs should also attempt to induce > 2 kW load reduction per home. Our analysis reveals that 90% of total superpeak period load drop in the summers of 2004 and 2005 was achieved by ADRS homes with ADU greater than 32 kWh; these homes made up 80% of the total high-consumption ADRS population.

In addition, we recommend that utilities use the following criteria to target future ADRS customers, in order to maximize future program performance:

- customers located in geographical subregions within the service territory that experience hottest summer temperatures, preferably above 90°F on average between 2 pm and 7 pm;
- customers who possess end uses in addition to A/C, such as swimming pool pumps and hot water heaters; and
- customers in regions where home construction and demographics resemble those of pilot participants in the SCE service territory: large, post-1985 homes that are more likely to have central air, and developments with higher-income households.

Also, we propose some guidelines for the design and implementation of future ADRS programs. These guidelines are intended to maximize load reduction, and therefore program effectiveness:

- Utilities should call superpeak event days when summer temperatures are highest (minimum of 90°F in regions for ADRS customers). Alternately, utilities should reserve as a separate category for event days called when

Average Reduction in Superpeak Consumption Relative to Control Homes, SCE High-Consumption ADRS Homes (2005)

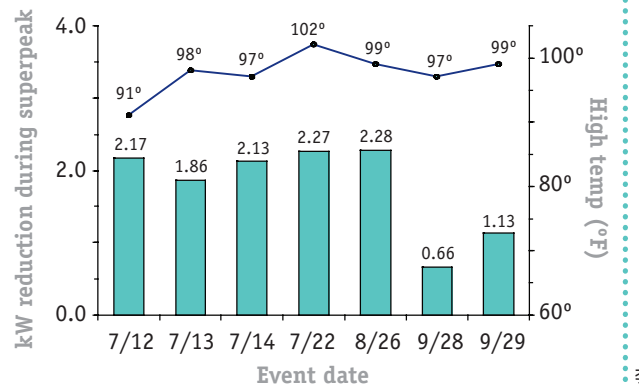


Figure 2. Event day averages show that, due to load reductions during the superpeak period, ADRS high-consumption homes consumed less energy than control homes during the superpeak period in both 2004 and 2005.

Moving Toward Further Reductions

Customers with ADRS technology and subject to dynamic, critical peak pricing rates in the inland (hot) climate zone successfully achieved load reductions compared to control customers without ADRS technology on standard tiered rates. The load reductions were substantial and stable across a range of days and temperatures. Technology appears to be an important driver in reducing load, especially peak period load, for homes with summer average daily usage greater than 24 kWh.

To improve the performance, and therefore the cost-effectiveness, of future ADRS programs, these programs should

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temperatures are merely warm or moderate, and should call event days independently rather than statewide.

- Utilities should shift the end of the peak period from 7 pm to 5:30 pm. Practically, utilities will probably want to stagger the end of the superpeak period to control the length of the recovery period. If all homes suddenly switch to off-peak mode at once, thermostats will revert to their off-peak settings and cause a large increase in consumption during the next two hours. This will create another system peak between 5:30 p.m. and 7 p.m.
- Utilities should shift the start of the peak period to 3 pm.
- Utilities should place ADRS customers on the CPP-V (day of rate instead of the CPP-F (day ahead) to maximize benefits, since the ADRS is automated.
- In limited situations, utilities should stagger calls to subsets of participants, rather than calling all participants at once, to even out the load reduction through the superpeak period.
- Utilities should call consecutive event days only when absolutely necessary. This will prevent “customer fatigue.”
- Utilities should employ ADRS technology as a load response program. This will render results more reliable, because they will be dispatched immediately.

Finally, we recommend that residential demand response programs for high-consumption households should include automated technology regardless of whether dynamic pricing is in place. In this way, utilities would have the ultimate flexibility to induce reductions in air conditioning and other residential end use loads in response to system needs. Automated technology could also improve price responsiveness in the absence of tariffs, or for customers who opt out of default dynamic tariffs.



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The authors gratefully acknowledge the contributions of L. Morton and E. Wanless to the original version of this document.